DIPLOMA PROGRAMME PRACTICE / HL / SET B

Economics / Paper 3

Worked mark scheme · 105 minutes · 60 marks

Answer all parts of both questions. A calculator is permitted. Write your answers and diagrams in the spaces provided. Show working and units. Give exact values or round to two decimal places unless otherwise instructed.

Judge extended responses as a whole using the markband for that part. Award valid alternative analysis and diagrams. In linked calculations, credit correct use of an earlier result. The suggested judgements are examples.

HOW TO MARK THIS PAPER

Use the analytic scheme in part (a). Apply the 10-mark policy scale to each part (b). Add the marks for both compulsory questions.

10-MARK SCALE

0: no relevant response. 1–2: a policy is named with minimal reasoning. 3–4: a plausible policy with partial explanation. 5–6: a relevant mechanism and some use of the case, with limited trade-offs. 7–8: developed theory, relevant data and reasoned consideration of limitations. 9–10: a well-supported, feasible recommendation synthesising case evidence, economic reasoning and the most consequential trade-offs. Credit a different justified recommendation.

QUESTIONS IN THIS SCHEME

1 A tariff on an imported staple [30]

2 A currency under pressure [30]

QUESTION 1 / 30 MARKS

A tariff on an imported staple

PART a(i) / 2 MARKS

Calculate the free-trade quantities supplied and demanded domestically.

Working. Qs = 20 + 5 × 12 = 80 thousand sacks (1). Qd = 200 − 5 × 12 = 140 thousand sacks per month (1).

PART a(ii) / 2 MARKS

Calculate monthly spending on grain imports under free trade.

Working. Imports = 140 − 80 = 60 thousand sacks (1). Spending = 60,000 × 12 = 720,000 dinars (1).

PART a(iii) / 3 MARKS

Calculate the level of imports after the tariff and the resulting monthly tariff revenue.

Working. The domestic price rises to 15, giving Qd = 125 and Qs = 95 thousand (1). Imports = 30 thousand (1). Revenue = 30,000 × 3 = 90,000 dinars per month (1).

PART a(iv) / 3 MARKS

Calculate the loss of consumer surplus per month caused by the tariff.

Working. Use the area between the two price lines and below demand (1). With linear demand, average quantity = (140 + 125)/2 = 132.5 thousand (1). Loss = 3 × 132.5 thousand = 397,500 dinars (1).

PART a(v) / 2 MARKS

Calculate the gain in domestic producer surplus per month.

Working. Average domestic supply over the price rise = (80 + 95)/2 thousand (1). Gain = 3 × 87,500 = 262,500 dinars (1).

PART a(vi) / 4 MARKS

Calculate the change in national welfare per month. Explain why counting only tariff revenue gives a misleading result.

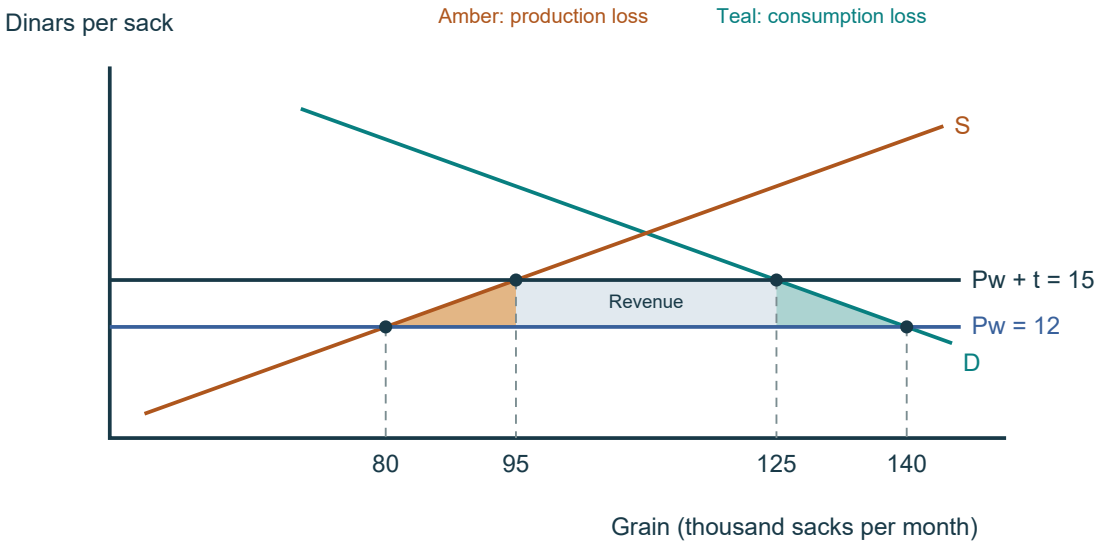
Working. Change = producer gain + revenue − consumer loss (1) = 262,500 + 90,000 − 397,500 = −45,000 dinars (1). Revenue is a transfer from buyers to the government (1). Extra high-cost domestic production and lost consumption account for the efficiency loss (1).

PART a(vii) / 4 MARKS

Using a tariff diagram, distinguish the part of the efficiency loss due to excess domestic production from that due to reduced consumption.

Diagram and explanation. Draw domestic demand and supply, with horizontal world-price and tariff-inclusive price lines. Mark the four quantities and the separate triangular areas (up to 2). Production loss = ½ × 3 × 15,000 = 22,500 dinars where domestic marginal cost exceeds world cost. Consumption loss = 22,500 where willingness to pay exceeds the world cost but purchases cease (up to 2).

WORKED REPRESENTATION / a(vii)



PART b / 10 MARKS

Using the information and your knowledge of economics, recommend a policy to support vulnerable grain farmers while protecting access to food in Lumen.

Worked recommendation. Use time-limited, means-tested farmer income support rather than raising the market grain price. The tariff transfers 262,500 dinars monthly to domestic producers but costs consumers 397,500 and creates a 45,000 efficiency loss. With grain taking 18% of the poorest households' budgets, a price rise can be especially burdensome. An income payment need not increase the marginal reward for uneconomic output.

Trade-offs. Targeting requires credible farm income records, general taxation has opportunity costs and large farmers may capture payments. Income relief does little for productivity. Storage support could help where losses are large but its value depends on reliable electricity. Set an expiry and monitor farm income, retail food prices and take-up by small farmers. Credit a defended tariff or storage recommendation that deals with these costs.

Apply the 10-mark scale on the first page to this response as a whole.

QUESTION 2 / 30 MARKS

A currency under pressure

PART a(i) / 2 MARKS

Calculate the goods balance and the services balance.

Working. Goods = 8 − 10 = −US$2 billion (1). Services = 3 − 2 = +US$1 billion (1).

PART a(ii) / 2 MARKS

Calculate the current-account balance.

Working. Net primary income = −0.9 and net secondary income = +0.3 billion. Combine with net goods and services of −1 (1). Balance = −US$1.6 billion (1).

PART a(iii) / 3 MARKS

Net private financing supplies US$1.0 billion. Calculate reserve use, ending reserves and months of import cover.

Working. Reserve use = 1.6 − 1.0 = US$0.6 billion (1). Ending reserves = 6 − 0.6 = US$5.4 billion (1). Monthly imports = 12/12 = US$1 billion, giving 5.4 months of cover (1).

PART a(iv) / 3 MARKS

Calculate the percentage rise in the local-currency price of a US dollar and the percentage fall in the dollar value of a local unit if the new rate is adopted.

Working. The dollar rises from 5 to 6, an increase of 1/5 × 100 = 20% (1). One local unit changes from US$0.20 to US$1/6 (1). Its dollar value falls by 16.67% (1). The two denominators differ.

PART a(v) / 2 MARKS

A firm owes US$4 million due immediately. Calculate the change in the local-currency value of this liability at the new rate.

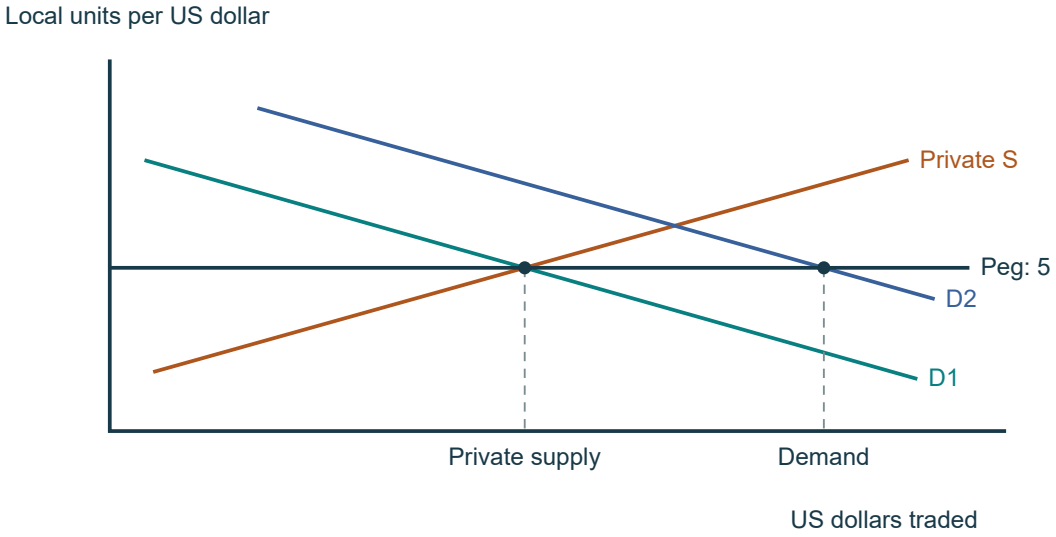
Working. The liability rises from 5 × 4 = 20 million to 6 × 4 = 24 million local units (1). Increase = 4 million local units (1).

PART a(vi) / 4 MARKS

Using a foreign exchange diagram for US dollars, explain how intervention can maintain the initial exchange rate when demand for dollars rises.

Diagram and explanation. Label local units per US dollar and quantity of dollars. Shift dollar demand right and show the excess demand at the original rate (up to 2). The central bank sells dollars for local currency to fill the gap, maintaining the rate while drawing down finite reserves (up to 2).

WORKED REPRESENTATION / a(vi)



PART a(vii) / 4 MARKS

Explain why lowering the value of the local currency might initially widen the trade deficit measured in local currency.

Explanation. With import volumes fixed by contracts, each dollar of imports costs more local currency. If export prices are fixed in local currency and capacity limits additional sales, local export receipts initially rise less than import expenditure. The later outcome depends on demand responses, capacity and currency of invoicing. The quotation alone does not determine the trade balance.

PART b / 10 MARKS

Using the information and your knowledge of economics, recommend an exchange-rate policy for Meridia in response to the continuing demand for dollars.

Worked recommendation. Allow a phased adjustment with limited intervention against abrupt movements. The annual funding gap of US$0.6 billion reduces cover from six to 5.4 months. Repeated defence without new financing would erode the buffer for essential imports. A weaker currency can curb imports and support exports over time.

Trade-offs. Immediate repricing raises the domestic cost of the illustrative dollar loan by 4 million local units; essential import demand is relatively inelastic. With tight transport capacity, rapid export expansion is unlikely. Aggressive defence avoids the immediate shock but can invite larger reserve losses if the rate is unsustainable. Publish conditions for intervention and monitor cover, debt service and essential prices. A temporary defence is defensible if the pressure is transient and financing is credible.

Apply the 10-mark scale on the first page to this response as a whole.